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Electronic vapor provision device

Abstract

An electronic vapor provision device comprising a power cell and a vaporizer, wherein the vaporizer comprises a heater and a heater support, wherein the heater is one the inside of the heater support.

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Background/Summary

RELATED APPLICATION (1) This application is a continuation of application Ser. No. 16/750,077, filed Jan. 23, 2020, which is a continuation of application Ser. No. 15/959,687, filed Apr. 23, 2018, now patented as U.S. Pat. No. 10,582,729, issued Mar. 10, 2022, which in turn is a divisional of application Ser. No. 14/415,540, filed Jan. 16, 2015, now patented as U.S. Pat. No. 9,974,335, issued May 22, 2018, which in turn is a National Phase entry of PCT Application No. PCT/EP2013/064950, filed Jul. 15, 2013, which claims the benefit of United Kingdom Application No. GB1212603.3, filed Jul. 16, 2012, each of which is hereby fully incorporated herein by reference.

TECHNICAL FIELD

(1) The specification relates to electronic vapor provision devices.

BACKGROUND

(2) Electronic vapor provision devices, such as electronic cigarettes, are typically cigarette-sized and typically function by allowing a user to inhale a nicotine vapor from a liquid store by applying a suction force to a mouthpiece. Some electronic vapor provision devices have an airflow sensor that activates when a user applies the suction force and causes a heater coil to heat up and vaporize the liquid.

SUMMARY

(3) In an embodiment there is provided an electronic vapor provision device comprising a power cell and a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support. One or more gaps may be provided between the heating element and the heating element support. Moreover, the electronic vapor provision device may have a mouthpiece section and the vaporizer may be part of the mouthpiece section. The heating element support may substantially fill the mouthpiece section.

(4) In another embodiment there is provided a vaporizer for use in the vapor provision device, that comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support.

(5) In another embodiment there is provided an electronic vapor provision device comprising a liquid store; a wicking element configured to wick liquid from the liquid store to a heating element for vaporizing the liquid; an air outlet for vaporized liquid from the heating element; and a heating element support, wherein the heating element is on the inside of the heating element support. The heating element support may be the wicking element. Moreover, the electronic vapor provision device may include a power cell for powering the heating element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the disclosure, and to show how example embodiments may be carried into effect, reference will now be made to the accompanying drawings in which:

(2) FIG. 1 is a side perspective view of an electronic cigarette.

(3) FIG. 2 is a schematic sectional view of an electronic cigarette having a parallel coil.

(4) FIG. 3 is a side perspective view of a heating element coil.

(5) FIG. 4 is a side perspective view of an outer heating element support.

(6) FIG. 5 is a side perspective view of a heating element coil within an outer heating element support.

(7) FIG. 6 is a side sectional view of a heating element coil within an outer heating element support.

(8) FIG. 7 is an end view of a heating element coil within an outer heating element support, where a central channel has a square cross-section.

- (9) FIG. 8 is an end view of a heating element coil within an outer heating element support, where a central channel has a circular cross-section.
- (10) FIG. 9 is an end view of a heating element coil within an outer heating element support, where a central channel has an octagonal cross-section.
- (11) FIG. 10 is an end view of a heating element coil within an outer heating element support having an outer square cross-section, where a central channel has a square cross-section.
- (12) FIG. 11 is an end view of a heating element coil within an outer heating support having two sections.
- (13) FIG. 12 is an end view of a heating element coil within a side channel of a heating element support.
- (14) FIG. 13 is an end view of a heating element coil within a side channel of a heating element support, with a second support section.
- (15) FIG. 14 is an end view of a heating element coil within a side channel of a heating element support having a rectangular cross-section.
- (16) FIG. 15 is an end view of a heating element coil within a side channel of a heating element support having a rectangular cross-section, with a second support section.

DETAILED DESCRIPTION

- (17) In an embodiment there is provided an electronic vapor provision device comprising a power cell and a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support. The electronic vapor provision device may be an electronic cigarette.
- (18) Having a separate heating element and support allows a finer heating element to be constructed. This is advantageous because a finer heating element can be more efficiently heated. Having the heating element on the inside of the support means that a much smaller and narrower heating element can be used since space is not needed inside the heating element to house a support. This enables a much larger and therefore stronger support to be used.
- (19) The heating element may not be supported on its inside. Having a heating element that is not supported on its inside means that a support does not interfere with the heating element on its inner region. This provides a greater heating element surface area which thereby increases the vaporization efficiency.
- (20) The heating element support may be a liquid store. A combined support and liquid store has the advantage that liquid can be easily transferred from the liquid store to the heating element supported by the liquid store. Also, by eliminating the need for a separate support, the device can be made smaller or a larger liquid store can be utilized for increased capacity.
- (21) One or more gaps may be provided between the heating element and the heating element support. Providing a gap between the heating element and the heating element support allows liquid to gather, and thereby be stored, in the gap region for vaporization. The gap can also act to wick liquid onto the heating element. Also, providing a gap between the heating element and support means that a greater surface area of the heating element is exposed thereby giving a greater surface area for heating and vaporization.
- (22) The heating element may be in contact with the heating element support at two or more locations. Moreover, the heating element may be in contact with the heating element support at points along the length of the support.
- (23) The heating element support may be a rigid and/or a solid support. Furthermore, the heating element support may be porous. For example, the heating element support may be formed of porous ceramic material.
- (24) The heating element support may be elongated in a lengthwise direction. Moreover, the heating element support may have a support channel and the heating element may be located in the support channel. Furthermore, the support channel may run in a lengthwise direction of the heating element support.

(25) The support channel may be an internal support channel. Moreover, the support channel may be a central support channel. Alternatively, the support channel may be a side support channel, located on a side of the heating element support.

(26) The support channel may be substantially cylindrical. Moreover, the cross-sectional shape of the support channel may be circular. Alternatively, the cross-sectional shape of the support channel may be a polygon. Furthermore, the cross-sectional shape of the support channel may have 4 sides, 6 sides or 8 sides. Cross-sections are sections perpendicular to the elongated lengthwise direction. These various shapes of support channel provide natural gaps between the support and a heating element coil within the support channel. These gaps lead to increased wicking, liquid storage and vaporization.

(27) The heating element support may comprise a first support section and a second support section. Moreover, the heating element may be supported by the first support section and the second support section. For example, the heating element may be supported between the first support section and the second support section. Furthermore, the support channel may be provided between the first support section and the second support section and the heating element may be in the support channel. The first support section may provide a first side of the support channel and the second support section may provide a second side of the support channel.

(28) Providing a support that comprises two separate sections provides an easier method of assembly. It also enables a more accurate and consistent positioning of the heating element relative to the support.

(29) The heating element may run along the length of the support channel. Moreover, the heating element may be in contact with the support channel at points along the length of the support channel. The heating element may be in contact with the surface of the support channel along the length of the support channel.

(30) The heating element may be a heating coil, such as a wire coil. The heating coil may be coiled so as to be supported along its length by the heating element support. The turns of the heating coil may be supported by the heating element support. The turns of the heating coil may be in contact with the heating element support. A gap may be provided between the heating coil and the heating element support. Moreover, the gap may be between a coil turn and heating element support. Furthermore, gaps may be between coil turns and the heating element support.

(31) By providing a gap between a coil turn and the support, liquid can be wicked into the gap and held in the gap for vaporization. In particular, liquid can be wicked by the spaces between coil turns and into the gap between a coil turn and the support.

(32) The vaporizer may have a vaporization cavity configured such that in use the vaporization cavity is a negative pressure region. At least part of the heating element may be inside the vaporization cavity, or the heating element may be entirely inside the vaporization cavity. For example, the vaporization cavity may be inside the heating element support. Moreover, the vaporization cavity may be inside a channel of the heating element support. At least part of the vaporization cavity may be inside the heating element.

(33) By having the heating element in the vaporization cavity, which in turn is a negative pressure region when a user inhales through the electronic vapor provision device, the liquid is directly vaporized and inhaled by the user.

(34) The electronic vapor provision device may further include a mouthpiece section and the vaporizer may be part of the mouthpiece section. Moreover, the heating element support may substantially fill the mouthpiece section.

(35) The liquid store may not comprise an outer liquid store container.

(36) Since the support is on the outside of the coil and can act as a liquid store, a liquid store container is not needed in addition to the liquid store, and the heating element support can fill the mouthpiece section to give greater storage capacity and a more efficient device.

(37) The electronic vapor provision device may further include a heating element connecting wire

and the heating element support may include a heating element connecting wire support section.

(38) The heating element support may be substantially cylindrical. The outer cross-sectional shape of the heating element support may be a circle. Alternatively, the outer cross-sectional shape of the heating element support may be a polygon. The outer cross-sectional shape of the heating element support may have 4 sides.

(39) Referring to FIG. 1 there is shown an embodiment of the electronic vapor provision device **1** in the form of an electronic cigarette **1** comprising a mouthpiece **2** and a body **3**. The electronic cigarette **1** is shaped like a conventional cigarette having a cylindrical shape. The mouthpiece **2** has an air outlet **4** and the electronic cigarette **1** is operated when a user places the mouthpiece **2** of the electronic cigarette **1** in their mouth and inhales, drawing air through the air outlet **4**. Both the mouthpiece **2** and body **3** are cylindrical and are configured to connect to each other coaxially so as to form the conventional cigarette shape.

(40) FIG. 2 shows an example of the electronic cigarette **1** of FIG. 1. The body **3** is referred to herein as a battery assembly **5**, and the mouthpiece **2** includes a liquid store **6** and a vaporizer **7**. The electronic cigarette **1** is shown in its assembled state, wherein the detachable parts **2**, **5** are connected. Liquid wicks from the liquid store **6** to the vaporizer **7**. The battery assembly **5** provides electrical power to the vaporizer **7** via mutual electrical contacts of the battery assembly **5** and the mouthpiece **2**. The vaporizer **7** vaporizes the wicked liquid and the vapor passes out of the air outlet **4**. The liquid may for example comprise a nicotine solution.

(41) The battery assembly **5** comprises a battery assembly casing **8**, a power cell **9**, electrical contacts **10** and a control circuit **11**.

(42) The battery assembly casing **8** comprises a hollow cylinder which is open at a first end **12**. For example, the battery assembly casing **8** may be plastic. The electrical contacts **10** are located at the first end **12** of the casing **8**, and the power cell **9** and control circuit **11** are located within the hollow of the casing **8**. The power cell **9** may for example be a Lithium Cell.

(43) The control circuit **11** includes an air pressure sensor **13** and a controller **14** and is powered by the power cell **9**. The controller **14** is configured to interface with the air pressure sensor **13** and to control provision of electrical power from the power cell **9** to the vaporizer **7**, via the electrical contacts **10**.

(44) The mouthpiece **2** further includes a mouthpiece casing **15** and electrical contacts **26**. The mouthpiece casing **15** comprises a hollow cylinder which is open at a first end **16**, with the air outlet **4** comprising a hole in the second end **17** of the casing **15**. The mouthpiece casing **15** also comprises an air inlet **27**, comprising a hole near the first end **16** of the casing **15**. For example, the mouthpiece casing may be formed of aluminum.

(45) The electrical contacts **26** are located at the first end of the casing **15**. Moreover, the first end **16** of the mouthpiece casing **15** is releasably connected to the first end **12** of the battery assembly casing **8**, such that the electrical contacts **26** of the mouthpiece **2** are electrically connected to the electrical contacts **10** of the battery assembly **5**. For example, the device **1** may be configured such that the mouthpiece casing **15** connects to the battery assembly casing **8** by a threaded connection.

(46) The liquid store **6** is situated within the hollow mouthpiece casing **15** towards the second end **17** of the casing **15**. The liquid store **6** comprises a cylindrical tube of porous material saturated in liquid. The outer circumference of the liquid store **6** matches the inner circumference of the mouthpiece casing **15**. The hollow of the liquid store **6** provides an air passageway **18**. For example, the porous material of the liquid store **6** may comprise foam, wherein the foam is substantially saturated in the liquid intended for vaporization.

(47) The vaporizer **7** comprises a vaporization cavity **19**, a heating element support **20** and a heating element **21**.

(48) The vaporization cavity **19** comprises a region within the hollow of the mouthpiece casing **15** in which liquid is vaporized. The heating element **21** and a portion **22** of the support **20** are situated within the vaporization cavity **19**.

(49) The heating element support **20** is configured to support the heating element **21** and to facilitate vaporization of liquid by the heating element **21**. The heating element support **20** is an outer support and is illustrated in FIGS. **4** to **7**. The support **20** comprises a hollow cylinder of rigid, porous material and is situated within the mouthpiece casing **15**, towards the first end **16** of the casing **15**, such that it abuts the liquid store **6**. The outer circumference of the support **20** matches the inner circumference of the mouthpiece casing **15**. The hollow of the support comprises a longitudinal, central channel **23** through the length of the support **20**. The channel **23** has a square cross-sectional shape, the cross-section being perpendicular to the longitudinal axis of the support.

(50) The support **20** acts as a wicking element, as it is configured to wick liquid in the direction **W** from the liquid store **6** of the mouthpiece **2** to the heating element **21**. For example, the porous material of the support **20** may be nickel foam, wherein the porosity of the foam is such that the described wicking occurs. Once liquid wicks **W** from the liquid store **6** to the support **20**, it is stored in the porous material of the support **20**. Thus, the support **20** is an extension of the liquid store **6**.

(51) The heating element **21** is formed of a single wire and comprises a heating element coil **24** and two leads **25**, as is illustrated in FIGS. **3**, **5**, **6** and **7**. For example, the heating element **21** may be formed of Nichrome. The coil **24** comprises a section of the wire where the wire is formed into a helix about an axis **A**. At either end of the coil **24**, the wire departs from its helical form to provide the leads **25**. The leads **25** are connected to the electrical contacts **26** and are thereby configured to route electrical power, provided by the power cell **9**, to the coil **24**.

(52) The wire of the coil **24** is approximately 0.12 mm in diameter. The coil is approximately 25 mm in length, has an internal diameter of approximately 1 mm and a helix pitch of approximately 420 micrometers. The void between the successive turns of the coil **24** is therefore approximately 300 micrometers.

(53) The coil **24** of the heating element **21** is located coaxially within the channel **23** of the support. The heating element coil **24** is thus coiled within the channel **23** of the heating element support **20**. Moreover, the axis **A** of the coil **24** is thus parallel to the cylindrical axis **B** of the mouthpiece casing **15** and the longitudinal axis **C** of the electronic cigarette **1**. Moreover, the device **1** is configured such that the axis **A** of the coil **24** is substantially parallel to airflow **F** through the device when a user sucks on the device. Use of the device **1** by a user is later described in more detail.

(54) The coil **24** is the same length as the support **20**, such that the ends of the coil **24** are flush with the ends of the support **20**. The outer diameter of the helix of the coil **24** is similar to the cross-sectional width of the channel **23**. As a result, the wire of the coil **24** is in contact with the surface **28** of the channel **23** and is thereby supported, facilitating maintenance of the shape of the coil **24**. Each turn of the coil is in contact with the surface **28** of the channel **23** at a contact point **29** on each of the four walls **28** of the channel **23**. The combination of the coil **24** and the support **20** provides a heating rod **30**, as illustrated in FIGS. **5**, **6** and **7**. The heating rod **30** is later described in more detail with reference to FIGS. **5**, **6** and **7**.

(55) The inner surface **28** of the support **20** provides a surface for liquid to wick onto the coil **24** at the points **29** of contact between the coil **24** and the channel **23** walls **28**. The inner surface **28** of the support **20** also provides surface area for exposing wicked liquid to the heat of the heating element **21**.

(56) There exists a continuous inner cavity **31** within the electronic cigarette **1** formed by the adjacent hollow interiors' of the mouthpiece casing **15** and the battery assembly casing **8**.

(57) In use, a user sucks on the second end **17** of the mouthpiece casing **15**. This causes a drop in the air pressure throughout the inner cavity **31** of the electronic cigarette **1**, particularly at the air outlet **4**.

(58) The pressure drop within the inner cavity **31** is detected by the pressure sensor **13**. In response to detection of the pressure drop by the pressure sensor **13**, the controller **14** triggers the provision

of power from the power cell **9** to the heating element **21** via the electrical contacts **10**, **26**. The coil of the heating element **21** therefore heats up. Once the coil **17** heats up, liquid in the vaporization cavity **19** is vaporized. In more detail, liquid on the coil **24** is vaporized, liquid on the inner surface **28** of the heating element support **20** is vaporized and liquid in the portions **22** of the support **20** which are in the immediate vicinity of the heating element **21** may be vaporized.

(59) The pressure drop within the inner cavity **31** also causes air from outside of the electronic cigarette **1** to be drawn, along route F, through the inner cavity from the air inlet **27** to the air outlet **4**. As air is drawn along route F, it passes through the vaporization cavity **19**, picking up vaporized liquid, and the air passageway **18**. The vaporized liquid is therefore conveyed along the air passageway **18** and out of the air outlet **4** to be inhaled by the user.

(60) As the air containing the vaporized liquid is conveyed to the air outlet **4**, some of the vapor may condense, producing a fine suspension of liquid droplets in the airflow. Moreover, movement of air through the vaporizer **7** as the user sucks on the mouthpiece **2** can lift fine droplets of liquid off of the heating element **21** and/or the heating element support **20**. The air passing out of the air outlet **4** may therefore comprise an aerosol of fine liquid droplets as well as vaporized liquid.

(61) With reference to FIGS. **5**, **6** and **7**, due to the cross-sectional shape of the channel, gaps **35** are formed between the inner surface **28** of the heating element support **20** and the coil **24**. In more detail, where the wire of the coil **24** passes between contact points **29**, a gap **35** is provided between the wire and the area of the inner surface **28** closest to the wire due to the wire substantially maintaining its helical form. The distance between the wire and the surface **28** at each gap **35** is in the range of 10 micrometers to 500 micrometers. The gaps **35** are configured to facilitate the wicking of liquid onto the coil **24** through capillary action at the gaps **35**. The gaps **35** also provide areas in which liquid can gather prior to vaporization, and thereby provide areas for liquid to be stored prior to vaporization. The gaps **35** also expose more of the coil **24** for increased vaporization in these areas.

(62) Many alternatives and variations are possible. For example, in embodiments, the electronic vapor provision device **1** may be configured such that the coil **24** is mounted perpendicular to a longitudinal axis C of the device. Moreover, FIGS. **8** to **15** show examples of different heating rod **30** configurations.

(63) FIG. **8** shows another example heating element support **20**. This is similar to the example above with the exception that the internal channel **23** has a circular cross-section rather than a square one. The coil **24** fits inside the channel **23** such that the coil turns are in contact with the channel walls **28**. There is greater contact between the coil **24** and the channel walls **28** than the example above, with the entire coil **24** generally in contact with the channel walls **28** rather than contact at given points **29**.

(64) This increase in contact area means that more liquid can be transferred to the full length of the coil rather than particular points **29**. However, since the coil **24** is generally in constant contact with the heating element support **20**, less of the coil surface area is exposed. So in use, when the coil **24** heats up, there will be less vaporization surface.

(65) These two examples show that a balance can be achieved between the amount of liquid on the coil **24** and the amount of vaporization surface exposed. This balance is varied by changing the amount of contact between the coil **24** and the channel **23** of the heating element support **20**.

(66) FIG. **9** shows an example where the amount of contact between the coil **24** and the channel **23** walls **28** lies between the examples shown in FIGS. **7** and **8**. In this example, the channel **23** has an octagonal cross-section rather than a circle or a square. As such, the coil **24** has coil turns which are generally in contact with the channel **23** of the heating element support **20** at 8 points **29** of contact. More gaps **35** are provided by the configuration of FIG. **9** than the configuration of FIGS. **3** to **7**. Moreover, the provided gaps **35** are smaller, leading to greater capillary action at the gaps.

(67) When compared to the channel **23** with the square cross-section, the increased contact, greater number of gaps **35** and smaller gap sizes all facilitate increased liquid transfer onto the coil **24**. The

increased exposed coil **24** surface compared to the channel **23** with the circular cross-section allows for more exposed vaporization surface for increased vaporization.

(68) In this way it can be seen that providing a heating element support **20** with an internal channel **23** having a regular polygon cross-section can be used to modify the amount of liquid transfer and the degree of vaporization by selecting the number of polygon sides. Thus, an optimum channel **23** cross-section can be selected.

(69) In the examples above, the heating element support **20** has a cylindrical shape and therefore the outer surface cross-sectional shape is circular. This shape is advantageous because the mouthpiece **2** section is also cylindrical so the heating element support **20** can be efficiently fitted into the mouthpiece **2** to minimize wasted space.

(70) Other outer surface cross-sectional shapes may for example be configured as shown in FIG. **10** having a heating element support **20** with a square outer cross-sectional shape.

(71) FIG. **11** shows a heating element support **20** comprising a first support section **36** and a second support section **37**. The heating element support **20** is generally cylindrical in shape and the first support section **36** and second support section **37** are half cylinders, with generally semi-circular cross-sections, which are joined together to form the cylindrical shape of the heating element support **20**.

(72) The first support section **36** and second support section **37** each comprise a side channel **38**, or groove **38**, running along their respective lengths, along the middle of their otherwise flat longitudinal faces. When the first support section **36** is joined to the second support section **37** to form the heating element support **20**, their respective side channels **38** together form the heating elements support **20** internal channel **23**.

(73) In this example, the combined side channels **28** form an internal channel **23** having a square cross-sectional shape. Thus, the side channels **28** are each rectangular in cross-section. As in the examples above, the coil **24** is situated within the heating element support **20** internal channel **23**. Having a heating element support **20** that comprises two separate parts **36**, **37** facilitates manufacture of this component. During manufacturing, the coil **24** can be fitted into the side channel **28** of the first support section **36**, and the second support section **37** can be placed on top to form the completed heating element support **20**.

(74) Other arrangements can also be considered to aid the construction of the heating element support **20** and coil **24** combination. FIG. **12** shows an example having a generally cylindrical heating element support **20** similar to that shown in FIG. **7**. However, the internal channel **23** is comprises a side channel **38** and the coil is thus not completely enclosed. The coil **24** can therefore be easily fitted into the open side channel **23**, **38**. Because the channel **23**, **38** is open, the coil **24** has coil turns that are in contact with the channel walls **28** at three points **29** of contact rather than four.

(75) FIG. **13** shows an example similar to that shown in FIG. **12** where the heating element support **20** of FIG. **12** is a first support section **36** and a second support section **37** is arranged such that it runs along the open channel **23**, **38**, plugging the open channel **38** and thereby closing it, and providing a combined arrangement similar to that shown in FIG. **7**. Thus the coil **24** is enclosed inside an internal combined channel **23** and the coil turns are in contact with the channel **23** at four points **29** of contact, three points **29** of contact with the first support section **36** and one point **29** of contact with the second support section **36**.

(76) FIG. **14** shows an example similar to that shown in FIG. **12** with the exception that the heating element support **20** has an outer rectangular cross-sectional shape. The coil **24** has coil turns having three points **29** of contact with the heating element support **20** channel **23**.

(77) FIG. **15** shows an example similar to that shown in FIG. **13** where a first support section **36** has an open side channel **38** and the coil **24** is fitted in this side channel. A second support section **37** is placed next to the first support section so that the coil **24** is enclosed between the support sections providing an arrangement similar to that shown in FIG. **10**. The coil **24** has coil turns with

four points **29** of contact with the heating element support **20** channel **23**, **38**, three with the first support section **36** and one with the second support section **37**. Once the first support section **36** and the second support section **37** are joined to form the support **20**, the formed support is substantially rectangular.

(78) The wire of the coil **24** is described above as being approximately 0.12 mm thick. However, other wire diameters are possible. For example, the diameter of the coil **24** wire may be in the range of 0.05 mm to 0.2 mm. Moreover, the coil **24** length may be different to that described above. For example, the coil **24** length may be in the range of 20 mm to 40 mm.

(79) The internal diameter of the coil **24** may be different to that described above. For example, the internal diameter of the coil **24** may be in the range of 0.5 mm to 2 mm.

(80) The pitch of the helical coil **24** may be different to that described above. For example, the pitch may be between 120 micrometers and 600 micrometers.

(81) Furthermore, although the distance of the voids between turns of the coil is described above as being approximately 300, different void distances are possible. For example, the void may be between 20 micrometers and 500 micrometers.

(82) The size of the gaps **35** may be different to that described above.

(83) In embodiments, the support **20** may be located partially or entirely within liquid store **6**. For example, the support **20** may be located coaxially within the tube of the liquid store **6**.

(84) An air pressure sensor **13** is described herein. In embodiments, an airflow sensor may be used to detect that a user is sucking on the device **1**.

(85) The heating element **21** is not restricted to having a uniform coil **24**. Moreover, in embodiments the coil **24** is described as being the same length as the support **20**. However, the coil **24** may be shorter in length than the support **20** and may therefore reside entirely within the bounds of the support **20**. Alternatively, the coil **24** may be longer than the support **20**.

(86) An electronic vapor provision device **1** comprising an electronic cigarette **1** is described herein. However, other types of electronic vapor provision device **1** are possible.

(87) Liquid may not be wicked and/or stored by the support **20** and could instead be wicked from the liquid store **6** to the coil and/or the inner surface **28** of the support **20** by a separate wicking element. In this case, the support **20** may not be porous.

(88) Internal support channels **23** with cross-sectional shapes other than those described could be used.

(89) The electronic vapor provision device **1** is not restricted to the sequence of components described and other sequences could be used such as the control circuit **11** being in the tip of the device **1** or the liquid store **6** being in the body **3** rather than the mouthpiece **2**.

(90) The electronic vapor provision device **1** of FIG. **2** is described as comprising two detachable parts, the mouthpiece **2** and the body **3**, comprising the battery assembly **5**. Alternatively, the device **1** may be configured such these parts **2**, **5** are combined into a single integrated unit. In other words, the mouthpiece **2** and the body **3** may not be detachable.

(91) Reference herein to a vaporization cavity **19** may be replaced by reference to a vaporization region.

(92) Although examples have been shown and described it will be appreciated by those skilled in the art that various changes and modifications might be made without departing from the scope of the invention.

(93) In order to address various issues and advance the art, the entirety of this disclosure shows by way of illustration various embodiments in which the claimed invention(s) may be practiced and provide for superior electronic vapor provision. The advantages and features of the disclosure are of a representative sample of embodiments only, and are not exhaustive and/or exclusive. They are presented only to assist in understanding and teach the claimed features. It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the claims or

limitations on equivalents to the claims, and that other embodiments may be utilized and modifications may be made without departing from the scope and/or spirit of the disclosure. Various embodiments may suitably comprise, consist of, or consist essentially of, various combinations of the disclosed elements, components, features, parts, steps, means, etc. In addition, the disclosure includes other inventions not presently claimed, but which may be claimed in future. Any feature of any embodiment can be used independently of, or in combination with, any other feature.

Claims

1. An electronic vapor provision device comprising a power cell and a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support, wherein one or more gaps are provided between the heating element and the inner surface of the heating element support, wherein the heating element is in contact with the heating element support at two or more locations, wherein the heating element support is elongated in a lengthwise direction, wherein the vaporizer further comprises a vaporization region configured such that in use the vaporization region is a negative pressure region, wherein at least part of the heating element is inside the vaporization region, wherein the heating element support is substantially cylindrical, wherein the outer cross-sectional shape of the heating element support is a circle, and wherein the heating element support includes a support channel in which the heating element is located, wherein the cross-sectional shape of the heating element support channel is a polygon.
2. The electronic vapor provision device of claim 1, wherein the electronic vapor provision device further comprises a mouthpiece section and the vaporizer is part of the mouthpiece section.
3. The electronic vapor provision device of claim 1, wherein the heating element is located such that the axis of the heating element is substantially parallel to airflow through the device when a user sucks on the device.
4. The electronic vapor provision device of claim 1, wherein the heating element is the same length as the heating element support, such that ends of the heating element are flush with ends of the heating element support.
5. The electronic vapor provision device of claim 1, wherein the distance between the heating element and the inner surface of the heating element support at each gap is in the range of 10 micrometers to 500 micrometers.
6. The electronic vapor provision device of claim 1, wherein the heating element is not supported on its inside.
7. The electronic vapor provision device of claim 1, wherein the heating element support is porous.
8. The electronic vapor provision device of claim 1, wherein the vaporization region is inside the heating element support.
9. A mouthpiece section adapted to detachably couple to a battery assembly of an electronic vapor provision device comprising a power cell, wherein the mouthpiece section comprises a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support, wherein one or more gaps are provided between the heating element and the inner surface of the heating element support, wherein the heating element is in contact with the heating element support at two or more locations, wherein the heating element support is elongated in a lengthwise direction, wherein the vaporizer further comprises a vaporization region configured such that in use the vaporization region is a negative pressure region, wherein at least part of the heating element is inside the vaporization region, wherein the heating element support is substantially cylindrical, wherein the outer cross-sectional shape of the heating element support is a circle, wherein the heating element

support includes a support channel in which the heating element is located, and wherein the cross-sectional shape of the heating element support channel is a polygon.
